

Cationic, hydrolysable side-chain regulated self-renewal brush copolymers for antibacterial and antifouling marine coatings

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ABSTRACT

Copolymers with quaternized poly(lauryl methacrylate)-b-poly(2-(*N,N*-dimethylamino) ethyl methacrylate) (PLMA-b-PDMAEMA) as the side chain, polycaprolactone (PCL) and siloxane bridged with hexamethylene diisocyanate (HDI) as the main chain were developed for antifouling marine coatings. With the hydrolysis of the quaternized PLMA-b-PDMAEMA and the degradation of PCL in seawater, the outmost surface of the coating peeled off by water shearing force. By exploring the co-effects of side chains, PCL and siloxane at varying contents, it was found that coatings with higher content of side chains tended to yield better surface hydrophilicity, faster surface self-renewal speed and lower protein adsorption amount. However, coatings with higher content of siloxane demonstrated stronger antibacterial abilities, even though they possessed less hydrophilicity and lower surface renewal speed. In-lab tests showed that the antibacterial rate of the coating can reach 99.7 % against *E. coli* and 99.6 % against *S. aureus*. And the lowest adsorption amount of bovine serum albumin was 36 $\mu\text{g}/\text{cm}^2$. Marine field tests revealed that most of the surface area of the properly designed coatings was clear of marine organisms during ~ 330 d of testing.

1. Introduction

With the development of marine industries, marine biofouling has become a serious problem [1,2]. For example, fouling organisms adhering to the hull surface increased the surface roughness of the ship, amplified the ship's navigational resistance, and consequently led to more fuel consumption and greenhouse gas emissions. Currently, antifouling coatings are still the most effective strategy to prevent the biofouling. With the ban of tributyltin (TBT) in 2008 by the International Maritime Organization, Cu-based coatings have been widely utilized in antifouling industry [3], but the continuous release of copper ions pose toxicity to marine organisms. Based on the requirement of environmental protection and durable antifouling effectiveness, non-toxic and environmental friendly antifouling coatings have become the mainstream [4,5].

To achieve the eco-friendly antifouling coatings, various methods were employed to improve the surface physicochemical properties. Besides the addition of ecofriendly antifoulants, other strategies such as the introduction of zwitterionic or amphiphilic moieties [5–7], fabrication of biomimetic microstructures [4,8,9] and biomimetic coatings [10,11], development of switchable surfaces [12,13], etc. were explored.

Polyzwitterions with equal number of positive and negative charges were extensively utilized as foulant-repellent materials to promote the antifouling ability of the marine coatings [6,14]. Currently, poly(2-dimethylaminoethyl methacrylate) (PDMAEMA) bearing abundant tertiary amines attracted extensive attention, as the tertiary amine groups can be quaternized to cations, subsequently, betainized to zwitterions. The quaternary ammonium based cationic polymers possessed highly efficient and broad-spectrum bactericidal abilities [15]. And quaternized and betainized PDMAEMA brushes exhibited antibacterial and antifouling capabilities based on killing-release mechanism [16,17]. Hence, the quaternary ammonium cations combined with the fouling-resistant zwitterions endowed the surface of a coating with antifouling and bactericidal properties. However, the quaternized and zwitterionic polymer based coatings have poor mechanical properties and underwater durability [18,19]. Because of the high content of ionic groups, these cationic/zwitterionic polymers can hardly be incorporated into organic substrates at high dosages due to the large difference in molecular polarity [14,20].

To reduce the strong interaction between zwitterionic/cationic PDMAEMA chains, zwitterionic polysulfobetaine methacrylate (PSBMA) was grafted to low surface energy polydimethylsiloxane (PDMS) [21] or

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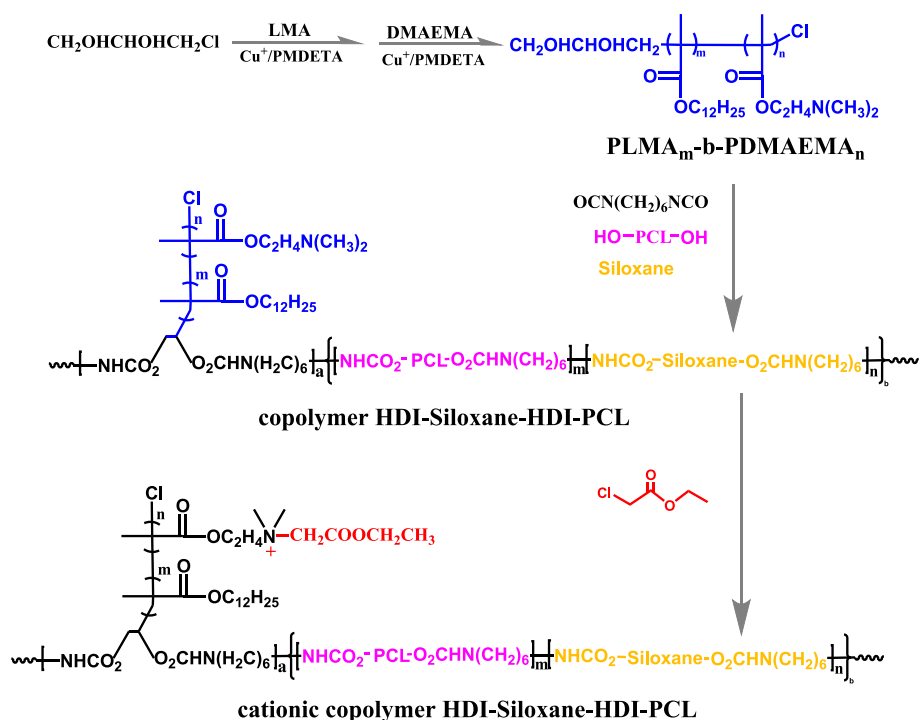
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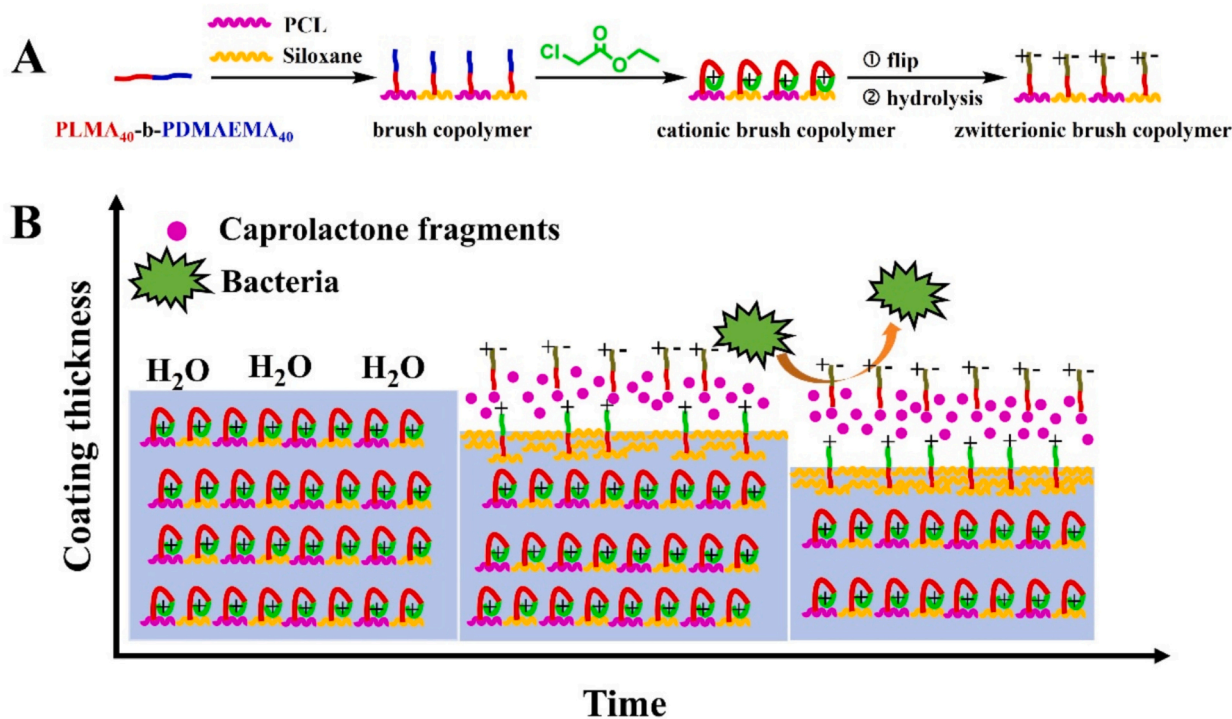
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Scheme 1. Synthetic route of cationic brush copolymers with PLMA-b-PCBMAE as the side chain and HDI-Siloxane-HDI-PCL as the main chain.



Scheme 2. Structure transformation (A) and antifouling mechanism (B) of the synthesized brush copolymers during hydrolysis in water.

PDMS-based polyurethane (PDMS-PU) [22] chains to acquire the anti-fouling coating with good mechanical properties. Moreover, amphiphilic copolymers aiming to mediate the polarity difference in the molecules were extensively investigated. For example, random-type amphiphilic zwitterionic copolymer consisting of carboxybetaine (CB) units and butyl methacrylate (BMA) units was developed for surface coating, its solubility in protic and aprotic solvent can be tuned by

changing the CB and BMA contents [23]. Amphiphilic block copolymers such as poly(lauryl methacrylate)-b-poly(2-(*N,N*-dimethylamino)ethyl methacrylate) (PLMA-b-PDMAEMA) [24] and amphiphilic cationic poly(hexadecyl methacrylate)-b-poly(carboxybetaine methacrylate ester) (PHMA-b-PCBMAE) [17] were synthesized to improve the compatibility of hydrophilic block in the coating, but the gelation of these copolymers at high blending percentages was still unavoidable. For example, the

Table 1

Molecular weight ratio of different compositions in the copolymer.

	PLMA ₄₀ -b-PDMAEMA ₄₀ (wt%)	Quaternization degree (wt%)	Siloxane (wt%)	PCL (wt%)	HDI (wt%)
Coating A	50	40	21.2	21.2	7.6
Coating B	50	40	28.3	14.1	7.6
Coating C	50	40	31.8	10.6	7.6
Coating D	25	80	31.9	31.9	11.2
Coating E	25	80	42.5	21.3	11.2
Coating F	25	80	47.9	15.9	11.2
Control 1			40	40	20
Control 2			53.3	26.7	20
Control 3			60	20	20

gelation of zwitterionic PSBMA and PDMAEMA copolymers occurred frequently [25].

Inspired by the utilization of amphiphilic block copolymer and the combination with low surface energy materials, brush copolymers with cationic poly(lauryl methacrylate)-b-poly(carboxybetaine methacrylate ester) (PLMA-b-PCBMAE) as the side chain was designed (Scheme 1). To fulfill the objective of sustainable antifouling, coatings with renewable surfaces are highly demanded. Hence, polycaprolactone (PCL) with polyurethane and polyester units which can degrade in the environment was synthesized as the main chain [26,27]. To decrease the high crystallinity of the PCL materials, siloxane moiety was introduced to the main chain as it is highly elastic, biocompatible and underwater stable [28]. The “hard” PCL segment and “soft” siloxane segment were linked by hexamethylene diisocyanate (HDI). With the cleavage of the C—O ester bond of PCL, the adhered foulants on coating surfaces together with degraded PCL fragments may be easily removed by water shearing force [22,27]. It was proved previously that the cationic amphiphilic

PLMA-b-PCBMAE copolymers tended to be self-assembled in organic solvent to form core-shell structure with the hydrophilic PCBMAE block as the inner core and the hydrophobic PLMA block as the outer shell, when in contact with water, the cationic amphiphilic copolymer switched its structure with cationic PCBMAE block as the outer shell [17,29]. Similarly, the cationic side chains of the synthesized copolymer could reach out to the seawater, with the hydrolysis of PCBMAE, the cationic structure transformed to zwitterionic (Scheme 2A). In other words, the brush-like side chains may form a hydration layer at the outmost surface of the coating, and HDI-Siloxane- HDI-PCL main chains may constitute the underneath layer. Therefore, the developed coating was expected to take effect in the synergistic antifouling and antibacterial functions (Scheme 2B).

With this design, copolymers composed of varied contents of cationic PLMA-b-PCBMAE side chains, PCL and siloxane were synthesized. Maximum quaternization degree of PLMA-b-PDMAEMA was tried with different contents of side chains. The positively charged side chains were

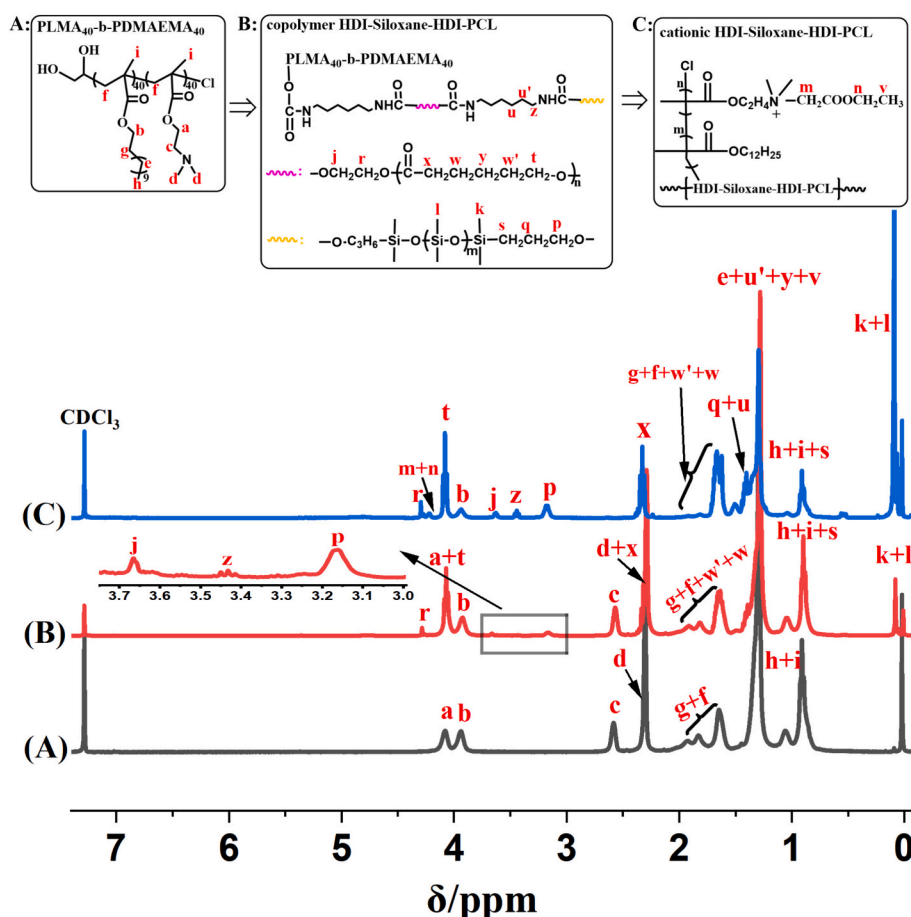


Fig. 1. ^1H NMR (in CDCl_3) spectra of polymer PLMA-b-PDMAEMA (A), copolymer HDI-Siloxane-HDI-PCL (B) and cationic copolymer HDI-Siloxane-HDI-PCL (C).

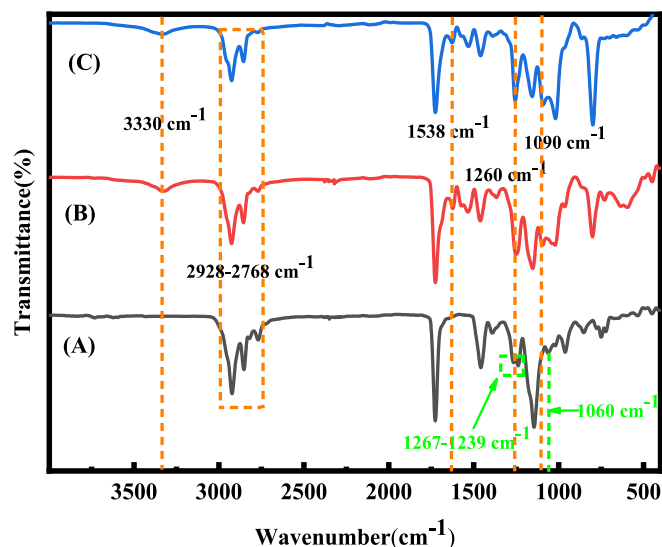


Fig. 2. FT-IR spectra of PLMA-b-PDMAEMA d (A), copolymer HDI-Siloxane-HDI-PCL (B) and cationic copolymer HDI-Siloxane-HDI-PCL (C).

in sterically repellent state, which may enhance the stability of the PLMA-b-PCBMAE copolymer, overcoming the gelation problem and promoting the antibacterial abilities of the coating [25,30].

Clearly, the designed copolymers are not only hydrolysable in the side chain but also degradable in the main chain. The continuous detachment of the outer layer of the coating guaranteed a self-renewal surface. The relationship between the designed molecular structure and the chemical/physical properties of the coating, which include the surface wettability, surface energy, surface renewal speed, mechanical properties, antifouling and antimicrobial performances, etc., were all investigated.

2. Experimental

2.1. Reagents

Dodecyl methacrylate (LMA), hexamethylene diisocyanate (HDI), ethyl chloroacetate, cuprous bromide (CuBr), bovine serum albumin (BSA), hydroxyl-terminated polycaprolactone (PCL, Mw 1000), hydroxypropyl-terminated siloxane (siloxane, Mw 1000), ethyl chloroacetate, dimethylaminoethyl methacrylate (DMAEMA), N,N,N',N'-pentamethyldiethylenetriamine (PMDETA) and tetrahydrofuran (THF) were all purchased from Aladdin (Aladdin Chemicals, Shanghai). Xylene, petroleum ether, 4,5-dichloro-2-octyl-isothiazolone (DCOIT) and 3-chloro-1,2-propanediol were purchased from Shanghai Jiuding

Chemical Co. Ltd. The control paint for marine field test was methacrylate based CSPC materials incorporated with 40 % Cu₂O (Kailin materials, Anhui).

2.2. One-pot synthesis of PLMA₄₀-b-PDMAEMA₄₀ diblock copolymers

Typical synthetic procedures of PLMA₄₀-b-PDMAEMA₄₀ were described previously [24] (Scheme 1). Firstly, CuBr (0.1435 g, 1 mmol) was added in a 250 mL three-neck flask, which was then degassed by nitrogen. Meanwhile, a mixture of 3-chloro-1,2-propanediol (0.1105 g, 1 mmol), PMDETA (0.3466 g, 2 mmol) and LMA (10.1763 g, 40 mmol) was bubbled with nitrogen in a test tube for 0.5 h, then introduced into the three-neck flask under nitrogen atmosphere. The resulting mixture was stirred at 40 °C for 12 h to obtain the macromolecular initiator PLMA-Cl. Afterwards, degassed DMAEMA (6.2885 g, 40 mmol) was added to the flask. The polymerization reaction was carried out at 80 °C for 24 h. The crude product was dissolved in THF, passed through an Al₂O₃ column and precipitated in methanol. The obtained PLMA₄₀-b-PDMAEMA₄₀ copolymer was verified via ¹H NMR spectrometry.

2.3. Synthesis and quaternization of brush copolymers

The synthetic route of the brush copolymer was described in Scheme 1. Typically, 1.5 g (0.0091 mM) PLMA₄₀-b-PDMAEMA₄₀, 1.91 g (1.91 mM) PCL, 0.65 g (3.87 mM) HDI, 1.91 g (1.91 mM) siloxane and 24.0 g

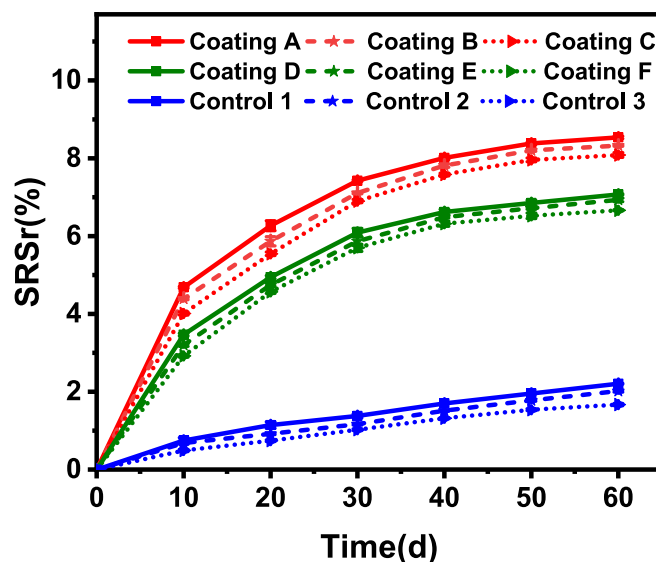


Fig. 4. Surface renewal speed of coatings in 60 d.

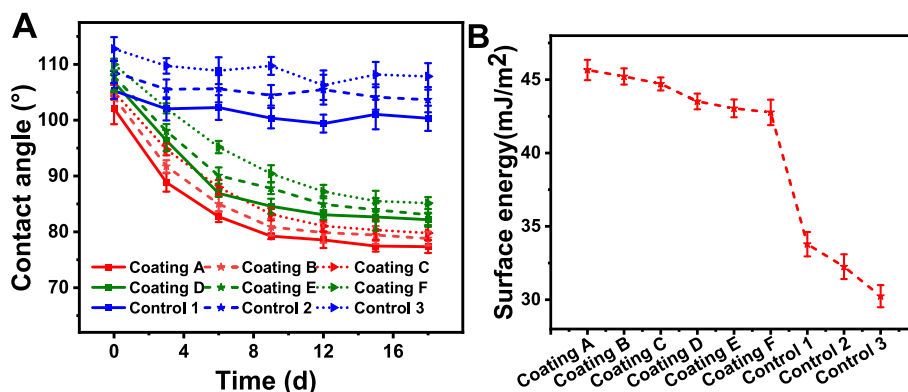


Fig. 3. Variations of WCA during hydrolysis (A) and surface energy after hydrolysis in seawater (B).

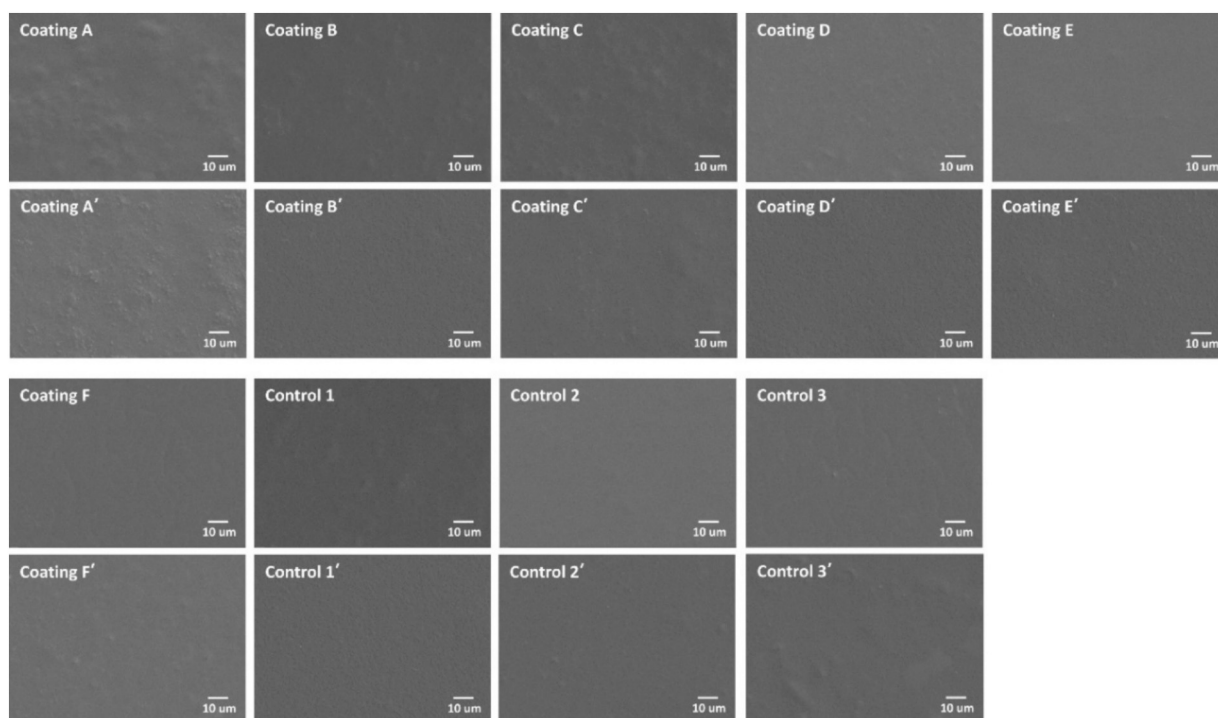


Fig. 5. SEM images of coating surfaces before hydrolysis (coatings A–F, controls 1–3) and after hydrolysis (coatings A'–F', controls 1'–3').

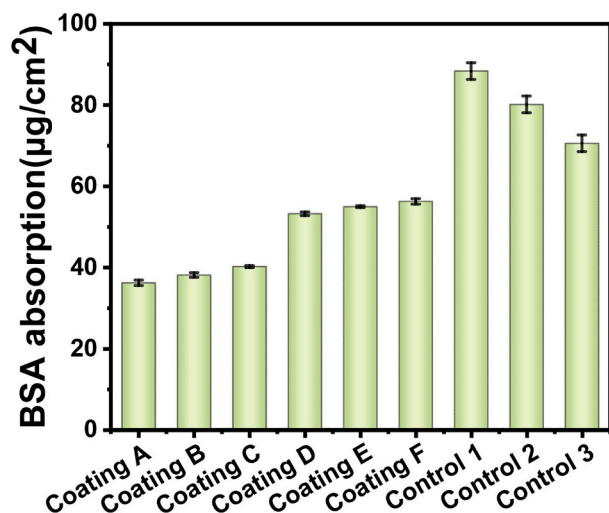


Fig. 6. The amount of BSA adsorption on coating surfaces.

xylylene were added in a round-bottom flask, the mixture was stirred at 60 °C for 6 h. The feeding ratio of the above procedure was according to the composition of coating D listed in Table 1. To measure the ^1H NMR spectrometry of the obtained product, a minute amount of the resulted mixture was precipitated, then filtered and evaporated. The obtained solids were dissolved in CDCl_3 for NMR testing.

Based on the above resulting product, the quaternization reaction was then performed by adding ethyl chloroacetate (0.36 g, 2.94 mM) to the above solution, and the mixture was refluxed at 60 °C for 30 h to obtain the cationic copolymer with a quaternization ratio of 80 % [17]. The resulted solution was precipitated in petroleum ether. After filtration and distillation under reduced pressure, the obtained cationic copolymer HDI-Siloxane-HDI-PCL was subjected to NMR testing.

To investigate the antifouling mechanism of different functional segments, a group of amphiphilic cationic block copolymers was

synthesized by adjusting the molar ratio of the monomers in the feeding process (Table 1). For coatings A, B and C with the listed compositions in Table 1, a maximum quaternization degree of 40 % PLMA-b-PDMAEMA side chain can be achieved, a higher degree than 40 % led to the gelation of the copolymer. And for coatings D, E and F, a maximum of 80 % quaternization can be reached. In addition to the quaternization degree, the contents of other components were also listed in Table 1, the given weight percentages represented the molecular weight ratio of the corresponding segment to that of the whole molecule. Controls 1, 2 and 3 with the same molecular weight ratio of siloxane:PCL as that of coatings A, B and C respectively were also synthesized.

The structures of the synthesized copolymers were characterized via ^1H NMR spectrometry (400 MHz, Bruker, Germany). The synthesized copolymers were also tested by Fourier transform infrared spectroscopy-attenuated total reflectance (FTIR-ATR, Thermo Electron Nicolet 380, USA) in the 4000–400 cm^{-1} wave number range.

2.4. Preparation and characterization of antifouling coatings

2.4.1. Coating preparation

A mixture containing 20 % (wt%) cationic copolymer and 80 % (wt %) xylene was coated on a PTFE plate. After natural drying at room temperature, the thickness of the coating was measured ($\sim 200 \mu\text{m}$). The prepared coatings were used for in-lab experimental tests.

2.4.2. Surface property test

Coatings were immersed in simulated seawater to hydrolyze for 18 d. The coatings were taken out to measure water contact angle (WCA) every three days on a goniometer (JGW-360 A goniometer, Chengde, China). The contact angles of water and diiodomethane were measured separately to obtain the surface free energy based on Owens-Wendt-Rabel-Kaelble two-liquid method [31], with each value representing the average of 5 repetitive tests on different locations of the same sample.

The prepared coatings were placed in a 2 % NaOH solution for 2 d to speed up the hydrolysis rate. Then, the coatings were dried and sprayed with gold. Surface morphology change of the coatings before and after

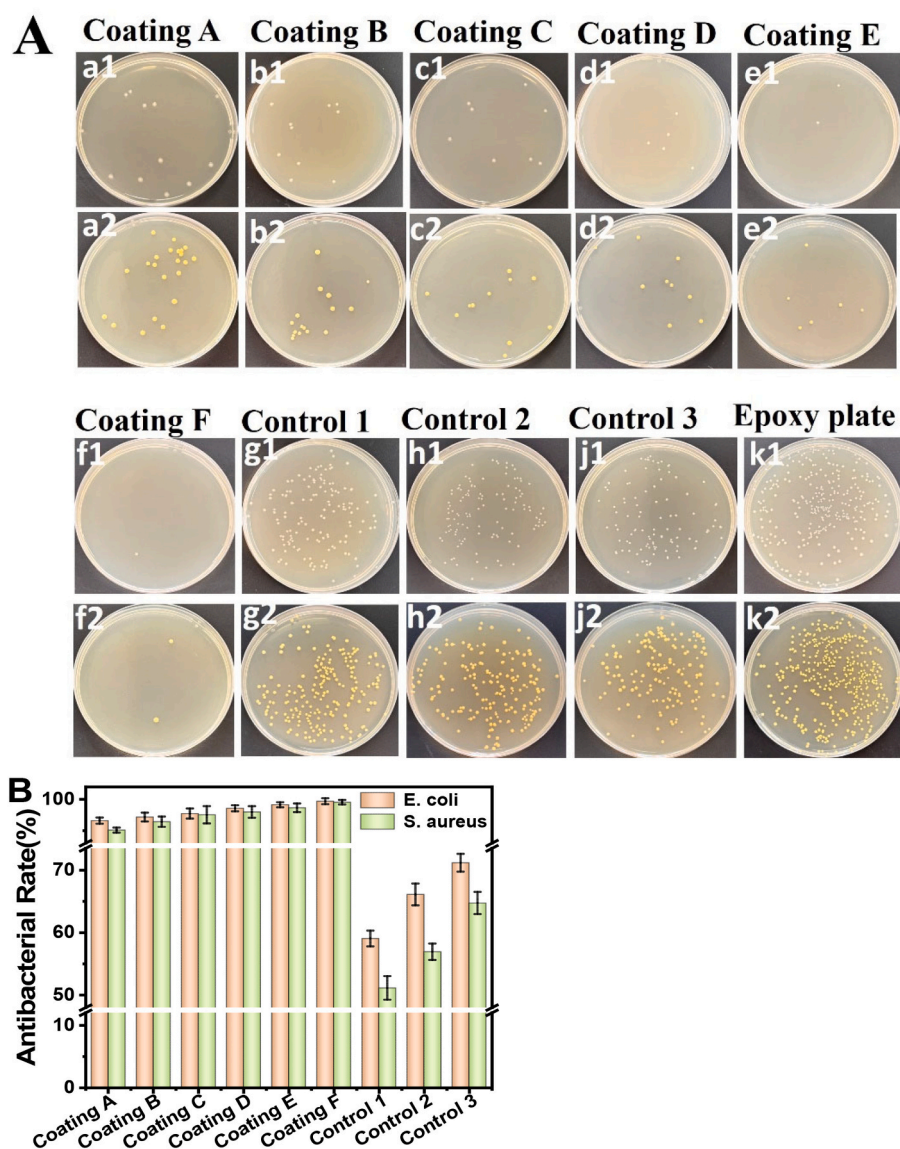


Fig. 7. Antimicrobial performances of different coatings against *E. coli* (a1–k1) and *S. aureus* (a2–k2) (A), and antibacterial rates of different coatings against *E. coli* and *S. aureus* (B).

hydrolysis was measured by field emission scanning electron microscope (SEM, ZEISS, Germany).

2.4.3. Surface renewal speed

About 2.0 g of coating solution with the concentration described in Section 2.4.1 was dropped on the surface of a 3 cm × 3 cm PTFE sheet, after natural drying, the coating was peeled off from PTFE plate and vacuum dried at 80 °C for 24 h. The coating was initially weighed as W_0 , and then immersed in seawater for 10 d. After flushing its surface to thoroughly remove the pollutants, the coating was vacuum dried and weighed again as W_1 . The coating was taken out every ten days to measure the weight loss during hydrolysis. The relative surface renewal speed (SRS, %) was calculated following the equation below:

$$SRS, \% = \frac{W_0 - W_1}{2W_0} \times 100\% \quad (1)$$

The presented value was the average of three replicate measurements.

2.4.4. Anti-protein adsorption

The prepared coating was cut into 2 cm × 2 cm pieces and immersed

in a fixed volume of PBS (pH = 7.4) with 1.0 g/L BSA (C_0). After adsorption for 24 h, the coatings were taken out to measure the amount of BSA adsorption on the surfaces ($\mu\text{g}/\text{cm}^2$) via Ultraviolet–visible (Uv-vis) spectroscopy at 280 nm (Beijing Purkinje General Instrument Co., Ltd.).

$$Q = \frac{(C_0 - C) \times V}{S} \times 1000 \quad (2)$$

Here, Q presents the measured membrane adsorption amount ($\mu\text{g}/\text{cm}^2$), C refers BSA concentration after adsorption (mg/mL), V (mL) is the volume of BSA solution and S is the area of all the tested films (cm^2). Each value was the average of triplicate measurements on different samples.

2.4.5. Antimicrobial test

Gram-negative *Escherichia coli* (*E. coli*) and Gram-positive *Staphylococcus aureus* (*S. aureus*) were selected to measure the antibacterial ability of the coating [32]. Briefly, the coating with a size of 1 cm × 1 cm was sterilized and placed in the bacteria solution with a concentration of 10^7 CFU/mL and cultured at 37 °C for 24 h. Next, the coating was taken out and washed with PBS to remove the non-adhered bacteria. The

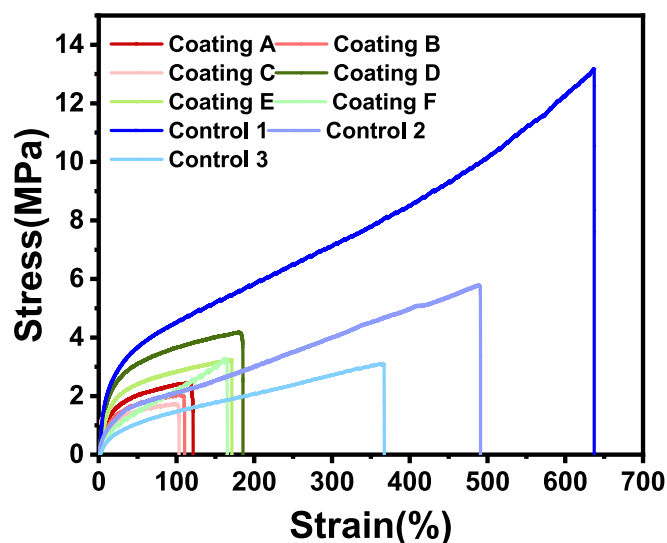


Fig. 8. Tensile stress-strain at break of different coatings.

bacteria left on the surface of the coating were then shaken off in a 5 mL PBS solution under sonication. After further diluted with PBS, 100 μ L of bacteria solution was transferred in the agar medium and cultured at 37 $^{\circ}$ C for 16 h. The number of colonies was then counted to evaluate the antibacterial rate. The antimicrobial activity of epoxy-coated blank plate was tested as well. The antibacterial rate of the coatings was calculated as follows:

$$\text{Antibacterial rate} = \frac{A - B}{A} \times 100\% \quad (3)$$

where, A represents the average colony number on the surface of epoxy-coated plate and B represents the colony number on the surface of other coatings. The above procedures were performed in triplicate with the same coating to calculate the average cell colonies.

2.4.6. Mechanical property test

The prepared coatings were cut into 4 mm $\times$ 45 mm rectangular pieces, and the tensile strength and elongation at break were measured with a 10 KN universal material testing machine (US Instron, US).

2.5. Marine field test

A mixture containing 20 % (wt%) cationic copolymer, 2 % (wt%) DCOIT and 78 % (wt%) xylene was coated on the surface of the epoxy plate. After natural drying, the plate was placed in a depth of 2–3 m below sea surface for field testing at Weihai (37 $^{\circ}$ 31'N, 122 $^{\circ}$ 01'E), Shandong Province from Aug 1st, 2023 to June 1st, 2024. The tested plates were taken out for observation every two weeks.

3. Results & discussion

3.1. Structural analysis

The three synthetic products showed in Fig. 1 were obtained via one-pot synthesis. The ^1H NMR spectrum of the product of atom transfer radical polymerization (ATRP) reaction was presented in Fig. 1A, where peak a at 4.06 ppm, c at 2.56 ppm and d at 2.28 ppm can be ascribed to $-\text{OCH}_2-$, $-\text{NCH}_2-$ and $-\text{NCH}_3$ groups of DMAEMA block, respectively. And characteristic peaks at 3.92 ppm (b), 1.62–1.90 ppm (f + g), 1.28 ppm (e) and 0.90–1.05 ppm (h + i) can be assigned to LMA block. All these characteristic peaks indicated the successful synthesis of PLMA-b-PDMAEMA diblock copolymer. Notably, the integrated area ratio of peak a to peak d was 1:3 in the spectrum A.

After the stepwise polymerization, peak r at 4.29 ppm corresponding to the $-\text{COOCH}_2-$ group of PCL segment appeared (Fig. 1B). And other characteristic peaks of PCL segment such as x at 2.32 ppm, w at 1.69 ppm, y at 1.29 ppm and w' at 1.62 ppm belonging to the hydrogen atoms of $-\text{O}_2\text{C}-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2-$ group respectively, were also shown in Fig. 1B. Additionally, the presence of peaks k + l at 0.06–0.09 ppm, peak s at 0.88 ppm and peak q at 1.44 ppm proved the existence of siloxane segment. And peak z at 3.42 ppm can be attributed to the hydrogen atom of methylene group connected to the secondary amine group of HDI segment. Therefore, the described ^1H NMR peaks in Fig. 1B were in good agreement with the predicted chemical compositions of copolymer HDI-Siloxane-HDI-PCL. Moreover, the integrated area ratio of peak a to peak d became 1:2 in Fig. 1B, suggesting that peaks t and x may have respectively overlapped with peaks a and d. However, after quaternization reaction, the ratio of the integrated area of these two overlapped peaks changed to 1:1 as shown in Fig. 1C.

The changes of integrated area ratio from 1:2 to 1:1 revealed that ^1H NMR signals of peaks a and d decreased after quaternization reaction (Fig. 1C). This phenomenon was due to the self-assembly behavior of cationic PCBMAE block as reported previously [17,29]. In this study, the partially quaternized DMAEMA block may be self-assembled and encapsulated by the hydrophobic PLMA block, which probably partially shield the ^1H NMR signal from PDMAEMA block. As a result, the hydrogen atoms of PLMA block still exist as shown in the spectrum of Fig. 1C, whereas the ^1H NMR signals from PDMAEMA block decreased significantly. Additionally, new peaks m + n at 4.21 ppm in accordance with the hydrogen atoms of $-\text{CH}_2\text{COOCH}_2-$ appeared in Fig. 1C, which are clearly introduced by ethyl chloroacetate. Unfortunately, ^1H NMR spectrum of cationic HDI-Siloxane-HDI-PCL copolymer in D_2O was not available due to its insolubility in water.

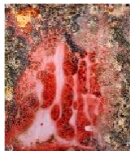
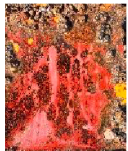
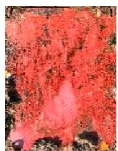
FTIR tests further illustrated the characteristic peaks of the synthesized polymers. As shown in Fig. 2A, the weak absorption at 1060 cm^{-1} and 1267–1239 cm^{-1} can be attributed to the bending vibration of hydroxyl end groups of PLMA-b-PDMAEMA copolymer [33]. And all three spectra showed absorption peaks at $\sim$ 2928–2768 cm^{-1} , which can be ascribed to the PDMAEMA block. Notably, the spectra in Fig. 2B and C exhibited similar absorption peaks, for example, both have N–H stretching vibration at 3330 cm^{-1} , N–H bending vibration at 1538 cm^{-1} of HDI segment, Si–C deformation bands at 1260 cm^{-1} and Si–O–Si asymmetric stretching vibration at 1090 cm^{-1} of siloxane segment [34]. Clearly, all these similar peaks in Fig. 2B and C are due to the same chemical groups of the two copolymers, even though HDI-Siloxane-HDI-PCL (Fig. 2C) is positively charged after quaternization reaction.

3.2. Surface properties

In addition to the degradation of PCL on the main chain, the cationic PCBMAE block on the side chain can also undergo hydrolysis in seawater, during which the self-assembled core-shell structure formed with cationic PCBMAE as core could transform its structure and create zwitterionic groups [17,24]. Hence, the hydrolysis of the cationic HDI-Siloxane-HDI-PCL copolymer may change the surface physical/chemical properties of the prepared coatings.

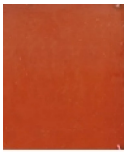
To investigate the hydrolysis behavior of the brush-like cationic copolymers in water, coatings A–F with varied contents of side chains, siloxane, and PCL were developed (Table 1). The PCBMAE block could enhance the hydrophilicity of the coatings, while the siloxane, PCL and LMA moieties could endow the coatings with hydrophobic nature. Therefore, the contradictory effects of these moieties may complex the surface properties of the coating. For controls 1–3, there were only slight differences in WCA between the three samples with the degradation of PCL in 18 d. For example, the WCAs were $\sim$ 100 $^{\circ}$ for control 1, $\sim$ 103 $^{\circ}$ for control 2 and $\sim$ 107 $^{\circ}$ for control 3. The difference in WCA of controls may stem from the different contents of siloxane (Fig. 3A). For coatings A–F, the surface hydrophilicity was largely promoted with the introduction of the cationic side chains. Before placing coatings in seawater,

Table 2
Pictures of marine field test in 330 d.

	0 d	60 d	150 d	240 d	330 d
Coating A					
Coating B					
Coating C					
Coating D					
Coating E					
Coating F					
Control 1					
Control 2					
Control 3					

(continued on next page)

Table 2 (continued)

	0 d	60 d	150 d	240 d	330 d
CSPC					
Blank epoxy plate					

it seemed that the initial WCA depended on the hydrophilic PDMAEMA and hydrophobic siloxane contents of the coating, which was coating A, B, C, D, E and F in an ascending sequence as shown in Fig. 3A. Within the first 12 d, coatings A–F displayed declining WCAs until a stable value was reached. And the final WCA value followed the same trend as that of the initial ones. Interestingly, the decreased WCAs of the six coatings were all $\sim 25^\circ$. It can be interpreted that the PCBMAE hydrolyzed and yielded zwitterionic groups, which enhanced the water-binding ability of the surfaces [24,35]. Besides, coatings A–F possess the same amount of cationic PCBMAEMA block, which is 20 % (wt%) of the whole molecules, therefore, coatings A–F demonstrated similar decreased WCAs. In fact, surface hydrophilicity is a combined effect of multiple composition of the copolymer, a comparatively higher content of PDMAEMA (50 %) and a lower content of siloxane (21.2 %) such as coating A would definitely yield better hydrophilicity than that of coating F with a PDMAEMA content of 25 % and a siloxane content of 47.9 %.

As the bioadhesion occurred at the interface of seawater and marine facilities, and it is well-known that both superhydrophilic surface [36] and superhydrophobic surfaces [37] may possess strong resistance to bioadhesion, hence, the surface energy of the coatings was explored. According to the Owens-Wendt-Rabel-Kaelble two-liquid method [38], coatings A–F exhibited similar surface energy ranging from 42 to 45 mJ/m² as shown in Fig. 3B, despite of different contents of siloxane in the cationic copolymers. In contrast, the control samples showed a distinctive lower surface energy of ~ 32 mJ/m², suggesting that the introduction of side chains remarkably upgraded the surface energy of coatings A–F. Clearly, the content of siloxane is not a key factor for surface free energy, as the coatings A–F with varied contents of siloxane from 21.2 % to 47.9 % have similar surface energy.

The above results evidenced that the introduction of cationic side chains promoted the surface wettability and the surface free energy, but the antifouling properties of the investigated coatings are still uncertain, as the surface energy of all the investigated coatings is higher than the minimum value of 22–24 mJ/m² illustrated by Baier curve [39,40].

3.3. Surface renewal speed

With the degradation of the main chain and the hydrolysis of the side chain, the polymeric debris remained on the surface of the coating can be flushed away by seawater. Moreover, tertiary amine and quaternized amine structure of the side chains can aid in the degradation reaction of PCL, thereby, accelerating the surface renewing process. Hence, even though the coating was in the static seawater, the weight loss rate still reflected the hydrolysis rate of the brush polymer, i.e., the self-renewal ability of the coating. It was depicted in Fig. 4 that the control coatings had a much slower surface renewal speed than the cationic brush copolymers, moreover, the surface renewal speed of coatings A, B and C was faster than that of the coatings D, E and F. Based on the above results, it was suspected that the hydrophilic side chain might be the

leading factor in determining the self-renewal speed of the coating, followed by the PCL. The mechanism might be that the PLMA-b-PCBMAE side chains are more hydrophilic than the main chain, therefore, cationic PCBMAE block would extend to the seawater while the main chain adhered to the coating surfaces via hydrophobic-hydrophobic interactions. With the hydrolysis of the side chains and the degradation of PCL, the hydrolyzed side chains together with the cleaved main chains could easily peel off by water shearing force. Hence, the surface renewal speed depends on the hydrolysis of the side chain and the degradation of the main chains in static water. Clearly, coatings A–F with PCL contents ranging from 15.9 % to 31.9 % demonstrated much faster self-renewal speed than that of the controls with PCL contents of 20 %–40 % (Fig. 4), indicating that the PLMA-b-PCBMAE side chain might be the decisive factor in determining surface renewal rate. For coatings A–C, the side chains account for 50 % of its whole molecular weight, however, the proportion decreases to only 25 % for coatings D–F. In view of the contents of side chains and PCL, it can be concluded that with similar molecular structure, higher content of side chains correlates with faster surface renewal speed, followed by higher content of PCL. Therefore, the self-renewal speed was in a descending order of coatings A, B, C, D, E, F, control 1, 2 and 3 as shown in Fig. 4.

3.4. Surface morphology

The surface morphology of the coatings before and after hydrolysis was tested and displayed in Fig. 5. It can be seen that except for coating A with the slightly rough surfaces probably owing to the less content of “soft” siloxane, all the other coatings showed smooth and uniform surfaces before and after hydrolysis. No obvious accumulation of polymeric debris or rough surfaces resulting from alkaline corrosion was observed. Clearly, with the hydrolysis of cationic PCBMAE and the degradation of PCL, siloxane fragments remained on coating surfaces were eroded gradually and finally washed away by water shearing force.

3.5. Antifouling and antimicrobial performances

Clearly, the zwitterionic groups generated by the cationic PLMA-b-PDMAEMA side chains (coatings A–F) are capable of protein resistance. To some extent, the low surface energy materials are also protein-repellent [41]. However, among the tested coatings, control 3 with the highest siloxane content of 60 % and the lowest surface energy of ~ 30 mJ/m² possessed limited antifouling ability. In contrast, coatings A, B and C demonstrated enhanced protein resistance capabilities compared to other coatings (Fig. 6). As for coatings A, B and C, the molecular weight of its hydrophilic side chains is approximately equal to that of the main chain, however, for coatings D–F, the molecular weight of the hydrophilic side chain is only half that of the main chain. In view of the surface hydrophilicity tests described in Section 3.2, i.e., more hydrophilic brushes lead to more complete coverage of antifouling groups on

coating surfaces [42], thereby, the protein adsorption amount on the surface of coatings A–C is less than that of coatings D–F. Additionally, among the coatings with the same amount of hydrophilic side chains, i. e., coatings A–C, coatings D–F, higher PCL content is in accord with more rapid surface replenishment, which favors the peeling-off the foulants on surfaces, consequently, protein adsorption resistance would be enhanced. Therefore, the protein resistance capability of the coatings followed the descending sequence $A > B > C > D > E > F$ as shown in Fig. 6.

Surfaces of coatings A–F and controls 1–3 were expected to have inhibitive effect on bacteria, as both zwitterionic groups resulted from the hydrolysis of PCBMAE and the siloxane segments in the copolymer are antibacterial [35]. Moreover, brushes on surfaces formed by the side chains are positively charged, electrostatic interactions between the polymer brush and cell wall of bacteria could cause the disruption of cell membranes and consequent death of cells [16,43]. Fig. 7A exhibited the densely populated bacteria on the blank epoxy plate. However, the number of cells significantly reduced on the agar surfaces of controls 1, 2 and 3. Obviously, control 3 has the strongest antibacterial ability among the three control samples, even though the self-renewal speed of control 3 is not as fast as the other two controls. This result suggested that for control samples, the siloxane in the polymer plays a much more important role in antibacterial performance than PCL, as control 1 contains 40 % PCL, whereas control 3 contains 20 % PCL. For the samples fabricated with cationic brush copolymers, coatings D–F exhibited better antibacterial performance than coatings A–C (Fig. 7B), even though coatings A–C have faster self-renewal speed. Judging by the molecular structure of the cationic copolymer and the control samples, it can be concluded that the content of siloxane in the copolymer is much more important in preventing the bacterial adhesion than the surface renewal speed. It can be explained that with the hydrolysis of PCBMAE and degradation of PCL, siloxane-PCL residuals may form a “slippery layer” to prevent the attachment of bacteria. And more siloxane moieties involved in the molecules of the copolymer mean more dense coverage of antibacterial fragments on surfaces (Scheme 2). In summary, the antibacterial activity of the hydrophilic brushes at the outmost layer of the coating, as well as the underneath siloxane layer constituted a protective structure to prevent the settlement of bacteria. Hence, for coatings prepared by cationic copolymers, the higher antibacterial rate correlates with lower surface free energy. Experimental results showed that the antibacterial rate of coating F was the highest among the tested coatings, which was 99.7 % against *E. coli* and 99.6 % against *S. aureus* (Fig. 7B). And coating D had a slightly higher antibacterial rate than coating C probably due to its comparatively higher PCL content.

3.6. Mechanical properties

Good mechanical properties such as higher tensile stress and strain at break can prevent the coatings from physical damages underwater. The test results in Fig. 8 illustrated that control samples especially control 1 exhibited an excellent tensile strength of 13.2 MPa and an elongation at break of 637 %, in contrast, the tensile strength and elongation at break of control 3 declined sharply (Fig. 8). It seemed that equal molecular weight of “soft” siloxane moiety and “hard” PCL moiety in the main chain endowed the coating favorable mechanical properties. Oppositely, a significant difference in molecular weight between siloxane and PCL decayed the mechanical properties of the coating. Further, the introduction of macromolecular side chain to the main chain inevitably undermined the mechanical properties of the polymeric materials [44], especially high content of side chains. For coatings A–C, the molecular weight of the side chain is equal to that of the main chain, therefore, the mechanical properties deteriorated more seriously compared with that of coatings D–F with molecular weight ratio of side chain to main chain of 1:3 (Fig. 8). Taking the contents of side chains, PCL and siloxane into consideration, the tensile stress and elongation at break of the coatings followed the descending order of D, E, F, A, B and C as shown in Fig. 8.

Despite of the poor mechanical properties of the materials with the introduction of side chains, the developed coatings especially D, E and F still have good mechanical properties for marine coatings.

3.7. Marine field tests

Coatings immersed in seawater for investigation of the settlement and growth of marine organisms on its surfaces could directly and effectively reflect the antifouling abilities of the coatings. Pictures in Table 2 demonstrated that most of the surface area on controls 1, 2 and 3 was covered with marine organisms in 60 d, and even worse, within the same time period, the blank epoxy-coated plate was completely occupied by flourishing marine organisms. In comparison, the commercially available CSPS coating was able to stand the marine foulants for ~240 d. Among the coatings fabricated with cationic copolymers, coatings A and B were partially covered by marine organisms in 60 d, in contrast, coatings C, D, E and F exhibited much stronger antifouling abilities. In the time duration of 330 d, the marine organisms still cannot habitat on the entire surfaces of coatings C, D, E and F. In particular, most of the surface areas of coatings E and F were free of bioadhesion probably due to the excellent antibacterial performances of the developed coatings. The results of marine field tests indicated that higher siloxane content in the cationic copolymer might be the key to achieve better antifouling performance.

4. Conclusions

Antifouling and antibacterial marine coatings were developed based on the cationic copolymers with PLMA-b-PCBMAE as the side chain and HDI-Siloxane-HDI-PCL as the main chain. The cationic/zwitterionic side chains were able to extend to the water and form hydrophilic brushes on the surface of the coating, while the main chain composed of siloxane and PCL constituted a low surface energy layer, which acted jointly to prohibit the adhesion of biofoulants. With the degradation of PCL in the main chain and the hydrolysis of the side chains, the degraded fragments peeled off from coating surfaces via water shearing force, avoiding the accumulation of microplastics on surfaces. Moreover, the continuous detachment of the outmost layer guaranteed a self-renewal surface, enhancing the antifouling and antibacterial ability of the coatings. The obtained results indicated that the content of hydrophilic PLMA-b-PCBMAE side chains may determine the self-renewal speed of the coating, followed by PCL, but the siloxane content decided the antibacterial rate of the cationic copolymers.

CRediT authorship contribution statement

Guilan Fu: Data curation. **Shixin Zhu:** Methodology, Formal analysis. **Xiuhua Sun:** Writing – original draft, Validation, Investigation. **Changlu Gao:** Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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